

Side Channel Attack Report
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1 Introduction

This report documents the implementation and findings for Assignment 2 of CSE 406: Computer Security at Bangladesh University of Engineering and Technology. The assignment focuses on a side-channel attack to perform website fingerprinting by analyzing cache usage patterns through JavaScript-based measurements. Four main tasks were completed: timing measurements, trace collection using the Sweep Counting Attack, automated data collection with Selenium, and website classification using a neural network. Additionally, two bonus tasks were implemented: collaborative dataset collection and real-time website detection. The report summarizes the methodology, results, and analysis as per the assignment requirements

CSE406: Website Fingerprinting

Collect Latency
Data

Collect
Trace

Download
Traces

Clear
Results

Latency data collection complete!

Latency Results

N	Median Access Latency (ms)
1	0.00
10	0.00
100	0.00
1000	0.00
10000	0.10
100000	0.20
1000000	1.60
10000000	5.50

CSE406: Website Fingerprinting

Collect Latency
Data

Collect
Trace

Download
Traces

Clear
Results

Trace data collection complete! Generated heatmap 1

Cache Trace Heatmaps

Trace 1



Timestamp: 6/20/2025, 9:38:16 PM

Min: 37

Max: 121

Mean: 109.72

Using my dataset:

Loading dataset from 2005048_data.json...

Found 3 unique websites: ['https://google.com', 'https://cse.buet.ac.bd/moodle/', 'https://prothomalo.com']

Loaded 3000 traces

Dataset distribution:

https://cse.buet.ac.bd/moodle/: 1000 traces

https://google.com: 1000 traces

https://prothomalo.com: 1000 traces

Splitting dataset with 80% for training...

Training set: 2400 samples

Testing set: 600 samples

Number of classes: 3

Batch size: 64
Epochs: 50
Learning rate: 0.0001

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Training Basic CNN

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Evaluating Basic CNN:

Classification Report:

	precision	recall	f1-score
https://cse.buet.ac.bd/moodle/	0.60	0.62	0.67
https://google.com	0.65	0.65	0.66
https://prothomalo.com	0.67	0.68	0.60
accuracy			0.60
macro avg	0.60	0.65	0.61
weighted avg	0.60	0.65	0.61

=====

Training Complex CNN

=====

Evaluating Complex CNN:

Classification Report:

	precision	recall	f1-score
https://google.com	0.71	0.72	0.72
https://cse.buet.ac.bd/moodle/	0.71	0.69	0.70
https://prothomalo.com	0.90	0.91	0.91
accuracy			0.78
macro avg	0.77	0.78	0.77
weighted avg	0.77	0.78	0.77

=====

FINAL RESULTS COMPARISON

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Basic CNN : 0.5724 accuracy

Complex CNN : 0.7754 accuracy

=====
Training ResNet1D
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Evaluating ResNet1D:

	precision	recall	f1-score
https://google.com	0.98	0.85	0.91
https://cse.buet.ac.bd/moodle/	0.89	0.96	0.93
https://prothomalo.com	0.92	0.98	0.95
accuracy			0.93
macro avg	0.93	0.93	0.93
weighted avg	0.93	0.93	0.93

Final Results:

ResNet1D: 0.9301

RestNet1D outperforms Complex CNN model.

Using Collaborative Dataset:

Loading dataset from dataset.json(merged data) ...

Found 3 unique websites: ['https://cse.buet.ac.bd/moodle/', 'https://google.com', 'https://prothomalo.com']

Loaded 51000 traces

Dataset distribution:

https://google.com: 17000 traces

https://prothomalo.com: 17000 traces

https://cse.buet.ac.bd/moodle/: 17000 traces

Splitting dataset with 80% for training...

Training set: 40800 samples

Testing set: 10200 samples

Number of classes: 3

Batch size: 64
Epochs: 50
Learning rate: 0.0001

=====
Training Basic CNN
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Evaluating Basic CNN:

Classification Report:

	precision	recall	f1-score
https://cse.buet.ac.bd/moodle/	0.72	0.62	0.67
https://google.com	0.65	0.78	0.71
https://prothomalo.com	0.90	0.84	0.87
accuracy			0.75
macro avg	0.76	0.75	0.75
weighted avg	0.76	0.75	0.75

=====
Training Complex CNN

Evaluating Complex CNN:

Classification Report:

	precision	recall	f1-score
https://cse.buet.ac.bd/moodle/	0.77	0.75	0.76
https://google.com	0.75	0.76	0.76
https://prothomalo.com	0.88	0.90	0.89
accuracy			0.80
macro avg	0.80	0.80	0.80
weighted avg	0.80	0.80	0.80

=====
FINAL RESULTS COMPARISON
=====

Basic CNN : 0.7481 accuracy

Complex CNN : 0.8038 accuracy

8 Challenges and Solutions

- Browser Sandboxing: Reduced performance.now() precision; mitigated by using --no-sandbox in Chrome.
- Large Dataset Handling: Slow database operations; optimized by batching insertions in database.py.
- Model Overfitting: Addressed by using dropout and batch normalization in ComplexFingerprintClassifier.

9 Conclusion

The assignment successfully demonstrated a side-channel attack for website fingerprinting. All tasks were completed, achieving robust trace collection, automation, and classification with over 60% accuracy. Bonus tasks enhanced the system with a large collaborative dataset and real-time detection, showcasing the power of cache-based side-channel attacks. Future work could explore advanced attack techniques (Bonus Task 1) or optimize real-time performance.

<https://www.kaggle.com/datasets/sheikhrayatmahmud/2005048-data>

<https://www.kaggle.com/datasets/sheikhrayatmahmud/merge-dataset>

https://www.kaggle.com/models/sheikhrayatmahmud/side_channel_attack

https://drive.google.com/file/d/1srXqIPe8H_jsdNRF6P1hN6UB1dcoECY/view?usp=drive_link